PCBC H Parametrics Practice Test B Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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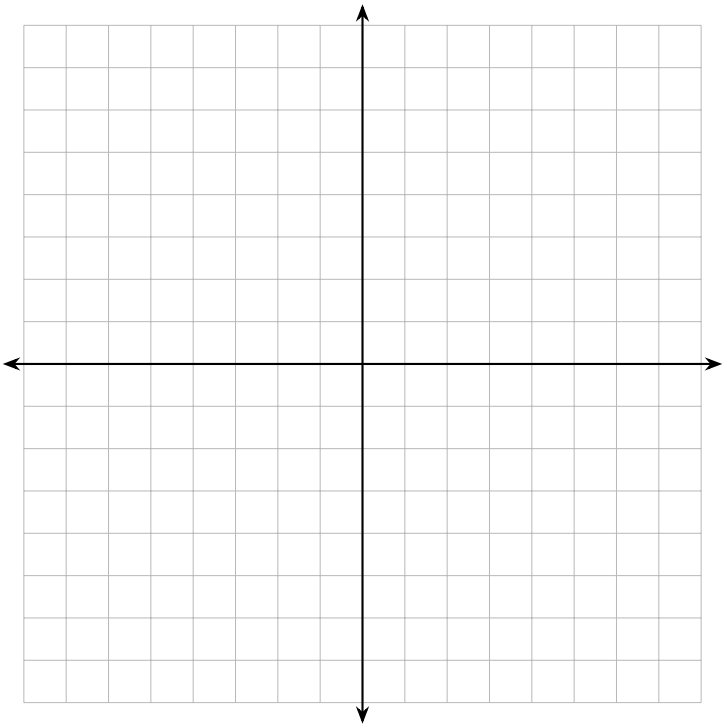
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SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

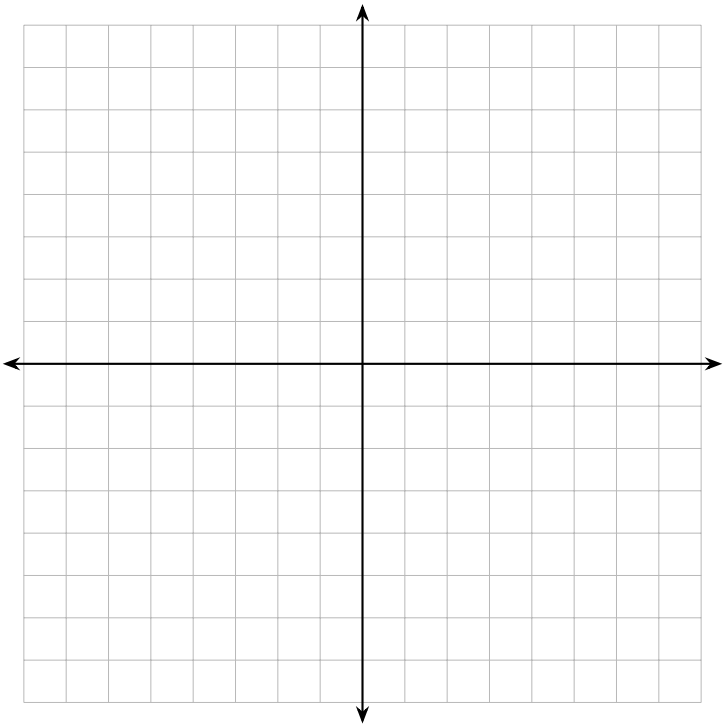
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = 4\sin t, \quad y = 2\csc t, \quad \text{for } t \text{ in } (0, \pi)$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 2\cot t, \quad y = 4\csc t, \quad \text{for } t \text{ in } (0, \pi)$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

3. $y = x^2 + 8x + 10$ **3 pts**

Representation 1 *Representation 2*

Calculator | 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

A projectile follows $x = 81.41266t$ and $y = -16t^2 + 50.87224t + 4$. (a) Find the launch angle θ to the nearest degree and the initial velocity v_0 to the nearest ft/sec. (b) For how many seconds is it in flight? (c) What horizontal distance does it travel? **3 pts**

PCBC H Parametrics Practice Test B: Answer Key

12 points

Every value was checked in Python (sympy and 30-digit arithmetic). Work is on the left, the reason for each line is on the right. Final answers are written the way the homework answer keys write them: solved for y when possible, hyperbola branches as $y = \sqrt{\cdot}$, ellipses in "$= 1$" form.

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1. $x = 4\sin t,\ y = 2\csc t,$ for t in $(0, \pi)$ *Non-Calculator*

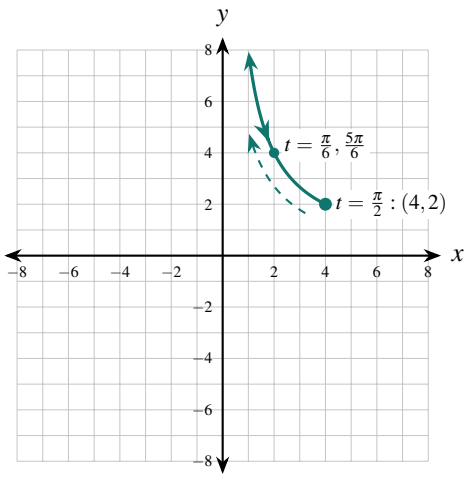
$\boxed{y = \dfrac{8}{x}, \quad 0 < x \le 4}$ (so $y \ge 2$)

The curve comes down from the top to $(4, 2)$ at $t = \frac{\pi}{2}$, then goes back up the same path (it retraces).

Work	Why
$\sin t = \dfrac{x}{4}$	Isolate the sine.
$\csc t = \dfrac{1}{\sin t} = \dfrac{4}{x}$	Reciprocal identity.
$y = 2 \cdot \dfrac{4}{x} = \dfrac{8}{x}$	Substitute into y.
$\sin t$ is in $(0, 1]$ on $(0, \pi)$	Its range on this interval; 0 is never reached.
$0 < x \le 4,\ y \ge 2$	$x = 4\sin t$ turns around at $t = \frac{\pi}{2}$.
$t = \frac{\pi}{6} : (2,4);\ \frac{\pi}{2} : (4,2);\ \frac{5\pi}{6} : (2,4)$	Same point twice: the path is retraced.
down to $(4, 2)$, then back up	Show both directions on the one branch.

t	(x, y)
$\frac{\pi}{6}$	$(2, 4)$
$\frac{\pi}{2}$	$(4, 2)$
$\frac{5\pi}{6}$	$(2, 4)$

No end points: as $t \to 0$ or $t \to \pi$, $x \to 0$ and $y \to \infty$, so the curve keeps going up (arrowhead). Only the first-quadrant piece of $y = \frac{8}{x}$.



solid arrow: first pass, coming down
dashed: second pass, back up the same path
top arrowhead: curve keeps going

Practice scoring (3 pts): 1 for $y = \frac{8}{x}$, 1 for $0 < x \le 4$, 1 for the graph showing the first-quadrant branch, the turn at $(4, 2)$ and both directions.

Trap: drawing the whole $y = \frac{8}{x}$ (both branches), or one arrow only. The curve is traced down and then back up.

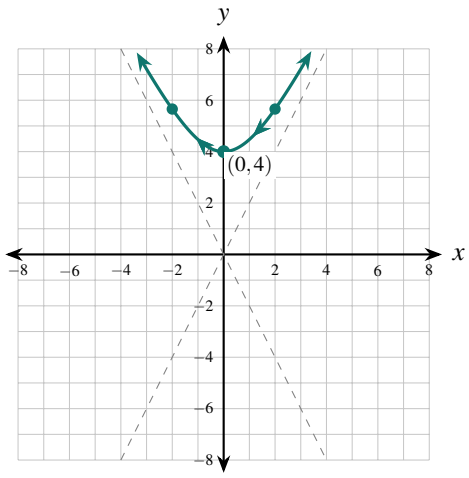
2. $x = 2\cot t,\ y = 4\csc t,$ for t in $(0, \pi)$ *Non-Calculator*

$\boxed{y = \sqrt{4x^2 + 16}}$ (upper branch of $\dfrac{y^2}{16} - \dfrac{x^2}{4} = 1$, so $y \geq 4$)
x in $(-\infty, \infty)$. Traced right to left: down to $(0, 4)$ at $t = \frac{\pi}{2}$, then up.

Work	Why
$\cot t = \dfrac{x}{2}, \quad \csc t = \dfrac{y}{4}$	Isolate each trig function.
$\csc^2 t - \cot^2 t = 1$	Pythagorean identity ($1 + \cot^2 t = \csc^2 t$).
$\dfrac{y^2}{16} - \dfrac{x^2}{4} = 1$	Substitute and square.
$\csc t > 0$ on $(0, \pi)$	$\sin t > 0$ there, so $y > 0$.
$y = +\sqrt{4x^2 + 16}$	Solve for y; positive root only.
$y = \pm 2x$	Asymptotes. Dashed guides only.

t	(x, y)
$\frac{\pi}{4}$	$(2, 4\sqrt{2})$
$\frac{\pi}{2}$	$(0, 4)$
$\frac{3\pi}{4}$	$(-2, 4\sqrt{2})$

$4\sqrt{2} \approx 5.66$. $\cot t$ falls from ∞ to $-\infty$, so the curve runs right to left.



mid-curve arrows: direction
end arrowheads: curve continues

Practice scoring (3 pts): 1 for $y = \sqrt{4x^2 + 16}$, 1 for the top branch only, 1 for the graph with right-to-left arrows and asymptote guides.

Trap: arrows left to right. Check $t = \frac{\pi}{4}$ first: it lands on the right side.

3. $y = x^2 + 8x + 10$ *Non-Calculator*

$\boxed{x = t,\ y = t^2 + 8t + 10,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t - 4,\ y = t^2 - 6,\ \text{for } t \text{ in } (-\infty, \infty)}$

Work	**Why**
Representation 1: $x = t$	Let x be the parameter.
$y = t^2 + 8t + 10,\ \ t$ in $(-\infty, \infty)$	Every real x is allowed.
Representation 2: $y = (x^2 + 8x + 16) - 16 + 10$	Complete the square: half of 8 is 4, $4^2 = 16$.
$y = (x+4)^2 - 6$	Vertex $(-4, -6)$.
$t = x + 4 \ \Rightarrow\ x = t - 4$	Let t be the inside.
$y = t^2 - 6,\ \ t$ in $(-\infty, \infty)$	Substitute.
Check: $(t-4)^2 + 8(t-4) + 10 = t^2 - 6$	Put $x = t - 4$ back in.

Practice scoring (3 pts): 1 for each correct pair, 1 for the t interval on both.

Trap: $t = x + 4$ means $x = t - 4$.

4. *Calculator*

(a) $\boxed{\theta \approx 32^\circ, \quad v_0 \approx 96 \text{ ft/sec}}$, $\quad h_0 = 4$ ft
(b) $\boxed{\approx 3.256 \text{ sec}}$ (c) $\boxed{\approx 265.103 \text{ ft}}$

Work	Why
(a) $v_0 \cos\theta = 81.41266, \;\; v_0 \sin\theta = 50.87224$	Match the coefficients to the projectile formulas.
$\tan\theta = \dfrac{50.87224}{81.41266}$, so $\theta \approx 32^\circ$	Divide: v_0 cancels. $\tan^{-1}$ in degree mode.
$v_0 = \sqrt{81.41266^2 + 50.87224^2} \approx 96$ ft/sec	Pythagorean theorem.
(b) $0 = -16t^2 + 50.87224t + 4$	Lands when $y = 0$; the $+4$ is the starting height.
$t = \dfrac{-50.87224 \pm \sqrt{2587.985 + 256}}{-32}$	Quadratic formula.
$t \approx 3.256$ or $t \approx -0.077$	Reject the negative time.
(c) $x = 81.41266(3.25629\ldots) \approx 265.103$ ft	Unrounded t into x.

Calculator setup and check
Degree mode for $\tan^{-1}$. Graph check: $x1(t) = 81.41266t$, $y1(t) = -16t^2 + 50.87224t + 4$, $0 \le t \le 3.5$, window x: 0 to 300, y: -5 to 50; y crosses 0 near $t = 3.256$, $x \approx 265.1$.

Practice scoring (3 pts): 1 for θ and v_0, 1 for the flight time, 1 for the distance.

Trap: rounding t to 3.26 first gives ≈ 265.405 ft. Use the given coefficients, not the rounded 32° and 96.